Earth Resource Satellites Systems

Earth Observation Systems Classification

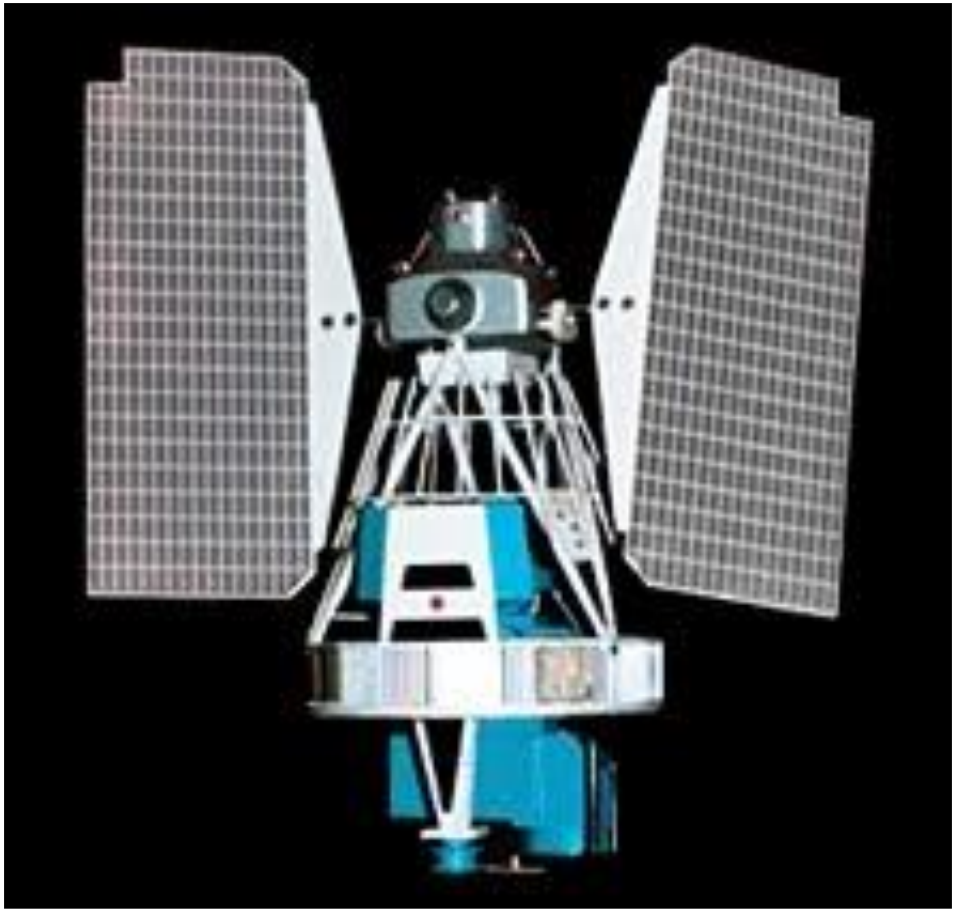
- **With respect to Spatial Resolution:**
  - **Very/ High spatial resolution: 0.6 - 4 m**
  - **Medium spatial resolution: >4 - 30 m**
  - **Low/coarse spatial resolution: >30 - > 1000 m**
- **With respect to Spectral resolution:**
  - **High spectral resolution: >220 bands**
  - **Medium spectral resolution: 3 - 15 bands**
  - **Low spectral resolution: < 3 bands**
- **With respect to Radiometric resolution:**
  - **Very/ High Radiometric resolution: > 8 bits**
  - **Medium Radiometric resolution: 6 - bits**
  - **Low Radiometric resolution: < 6 bits**

Earth Observation Systems Classification

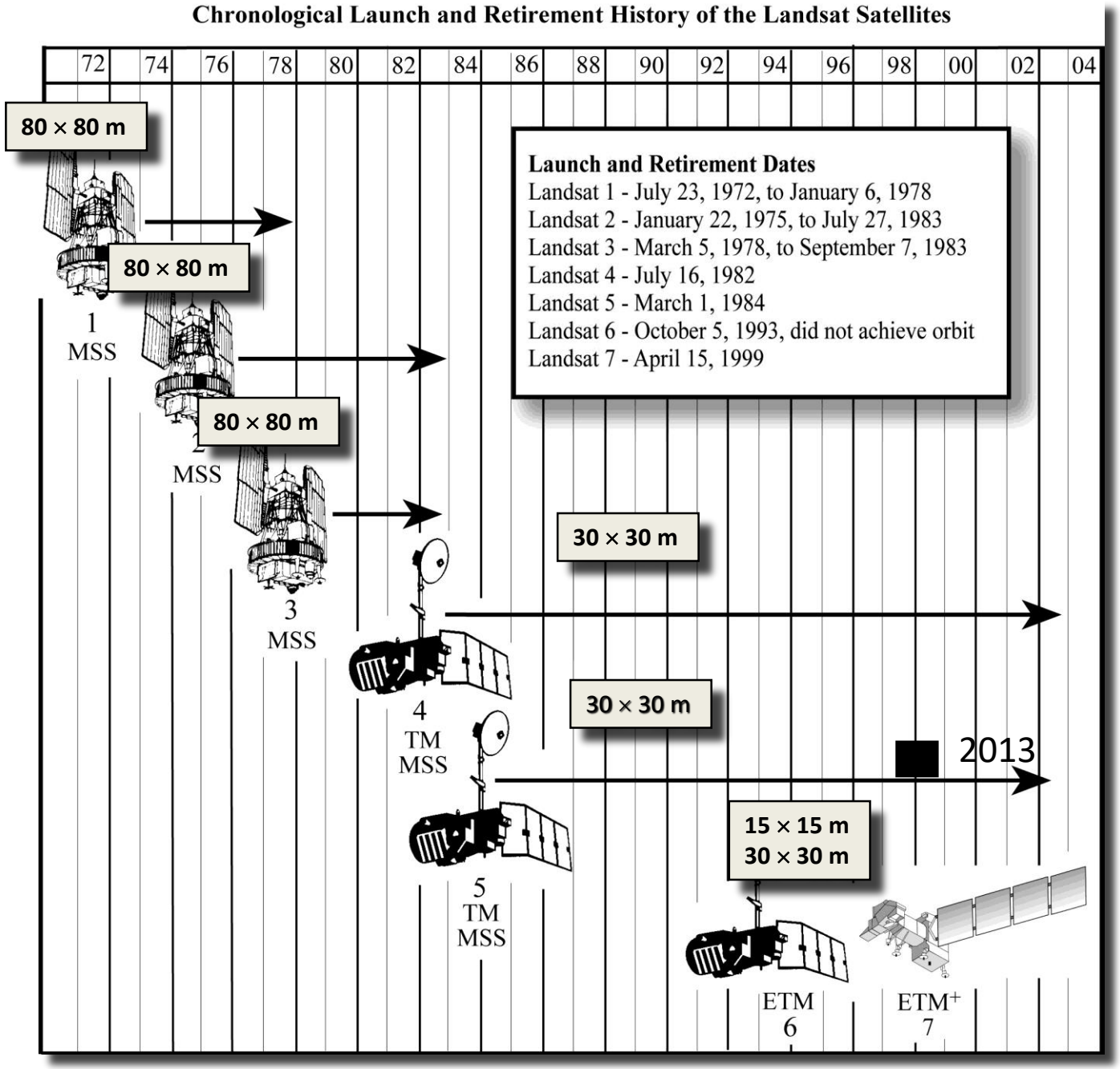
- **With respect to Temporal Resolution:**
  - **Very/ High temporal resolution:** < 24 hours - 3 days
  - **Medium temporal resolution:** 4 - 16 days
  - **Low temporal resolution:** > 16 days

MEDIUM RESOLUTION

LANDSAT



Landsat-1 originally called " Earth Resources Technology Sat-1"



LANDSAT CLASSIFICATION

- **Technically / based on sensors, Landsat sat can be classified into three groups:-**
  - **Landsat 1,2,3 -MSS**
  - **Landsat 4,5 - TM**
  - **Landsat 6,7- ETM, ETM+**
  - **Landsat 8 Operational Land Imager (OLI)**

LANDSAT 7 (ETM+)

(Enhanced Thematic Mapper)

- Specification same as TM 4,5,Except
- 8 bands Resolution
  - 6 Multispectral - 30 M
  - Thermal -60 m
  - Panchromatic - 15 M
- Equator Crossing- 10:00 AM

TM BANDS

Channel	Wavelength Range (mm)	Application
TM 1	0.45 - 0.52 (blue)	soil/vegetation discrimination;
TM 2	0.52 - 0.60 (green)	green vegetation mapping
TM 3	0.63 - 0.69 (red)	vegetated vs. non-vegetated ,cultural/urban feature identification
TM 4	0.76 - 0.90 (near IR)	identification of plant/vegetation types, health, and water body delineation; soil moisture
TM 5	1.55 - 1.75 (short wave IR)	sensitive to moisture in soil and vegetation; discriminating snow and cloud-covered areas
TM 6	10.4 - 12.5 (thermal IR)	Thermal mapping (urban, water)
TM 7	2.08 - 2.35 (short wave IR)	
ETM -8 Panchromatic (0.52-0.9)		

LANDSAT-7

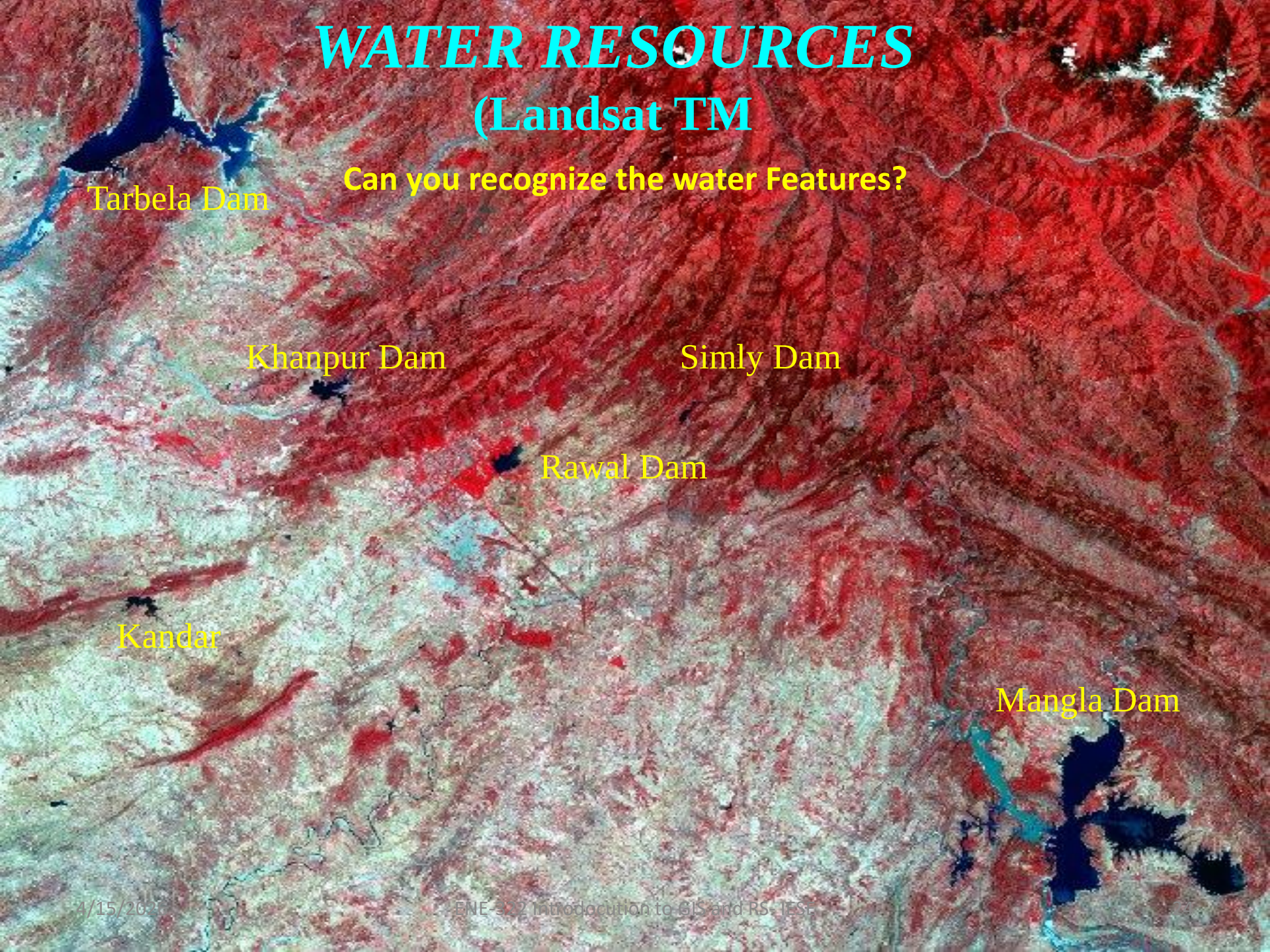
Scan line problem



WATER RESOURCES

(Landsat TM

Can you recognize the water Features?





Wildlife refuge pond in Hawaii mysteriously turns bright pink

An algae bloom, but lab tests found Instead an organism called halobacteria might be the culprit.



Halobacteria flourish in bodies of water with high levels of salt.

LANDSAT 8

(Landsat Data Continuity Mission (LDCM))

- Launched on February 11, 2013
- Operational Land Imager (OLI) and the Thermal Infra Red Sensor (TIRS)
- Resolution
  - Panchromatic - 15 M
  - Multispectral - 30 M
  - Thermal - 100 m
- Equator Crossing- 10:00 AM
- 12-bit quantization of data
- Pushbroom sensor

The Landsat 8 satellite images the entire Earth every 16 days in an 8-day offset from Landsat 7

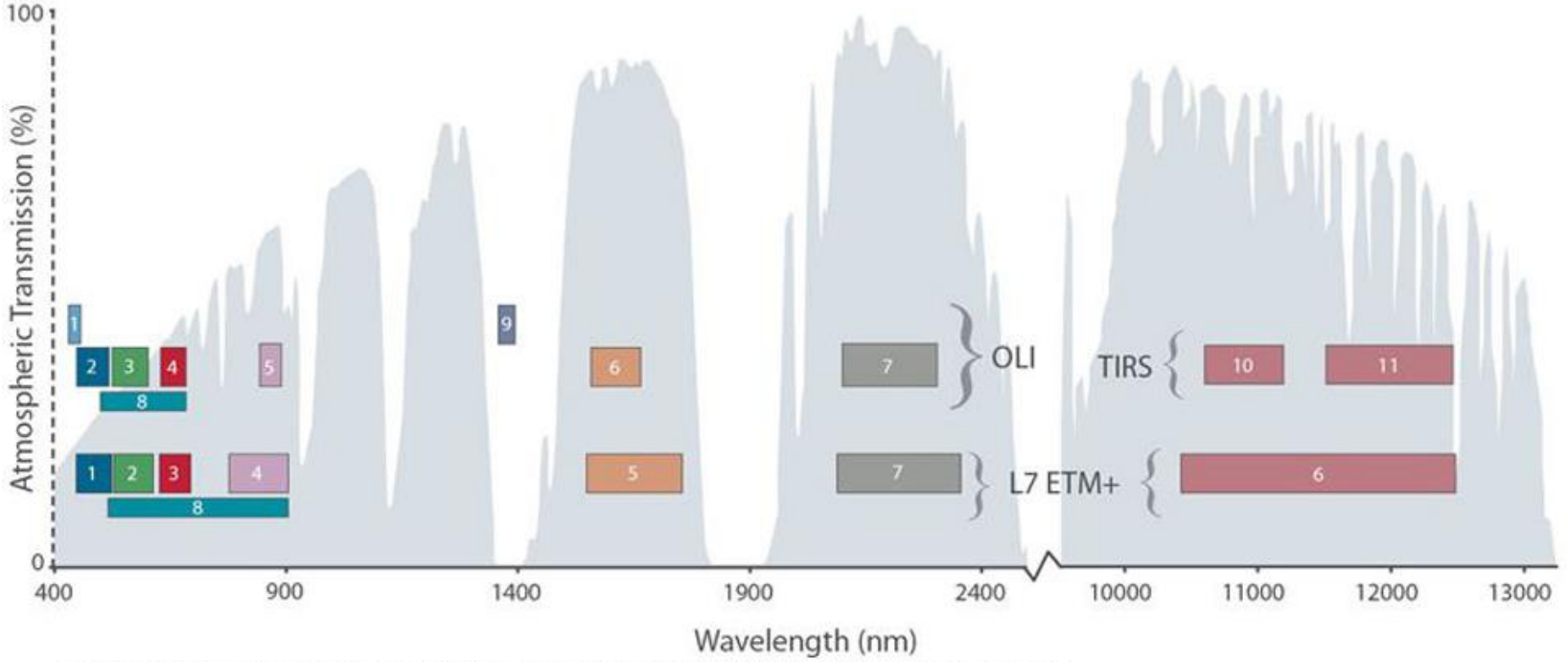
LANDSAT 8

(Landsat Data Continuity Mission (LDCM))

- **Orbit:** Sun-synchronous near-circular orbit
- **Altitude:** 705 km
- **Inclination :** 98.2º
- **Period :** 99 minutes
- **Repeat coverage :** 16 days (233 orbits)
- **Equator crossing time:** 10:00 AM

https://directory.eoportal.org/web/eoportal/satellite-missions/l/landsat-8-ldcm

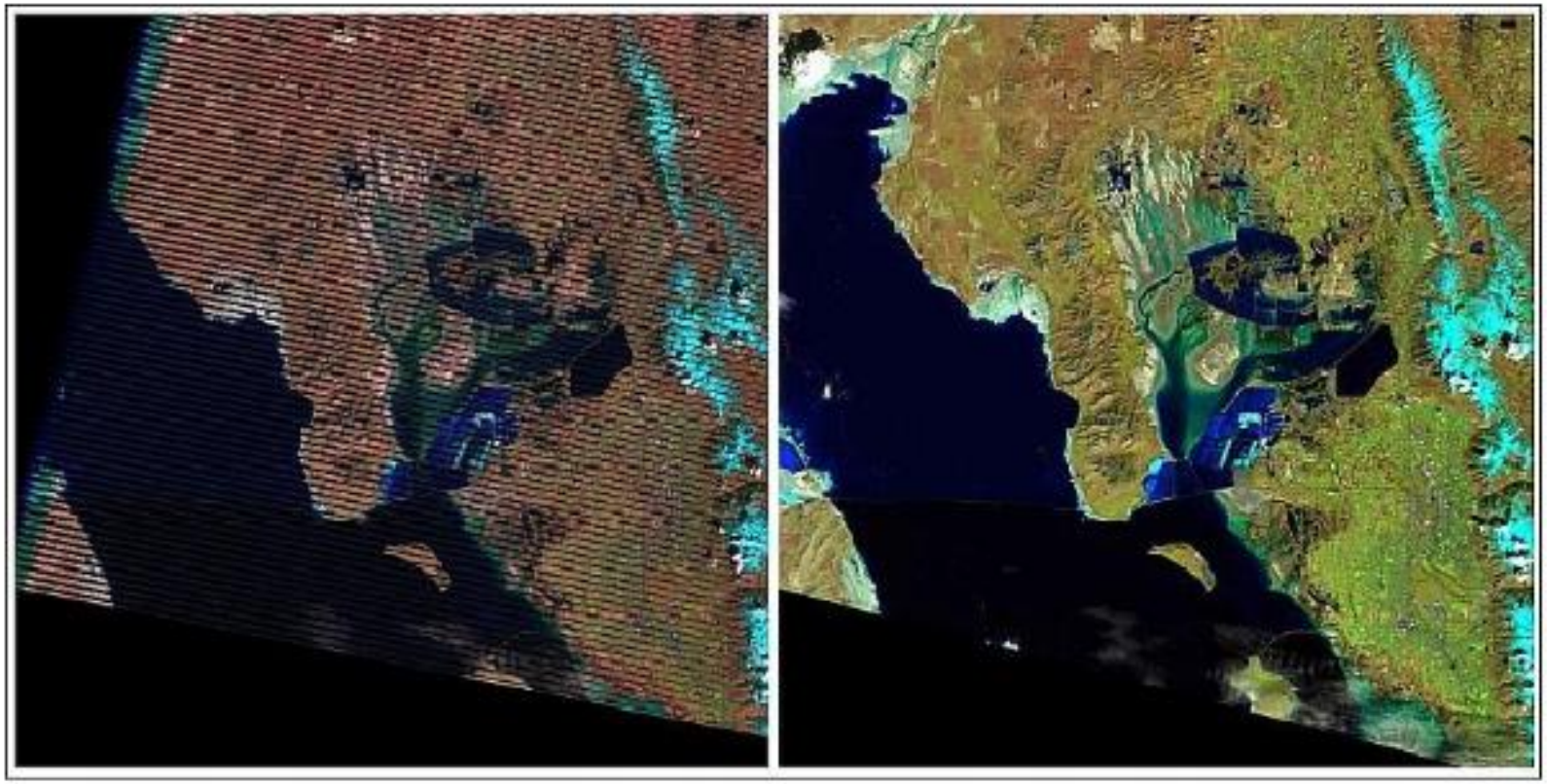
Landsat 8



Bandpass wavelengths for Landsat 8 OLI and TIRS sensor, compared to Landsat 7 ETM+ sensor

LANDSAT 7 Vs LANDSAT 8

(Landsat Data Continuity Mission (LDCM))



Landsat-7 Landsat-8

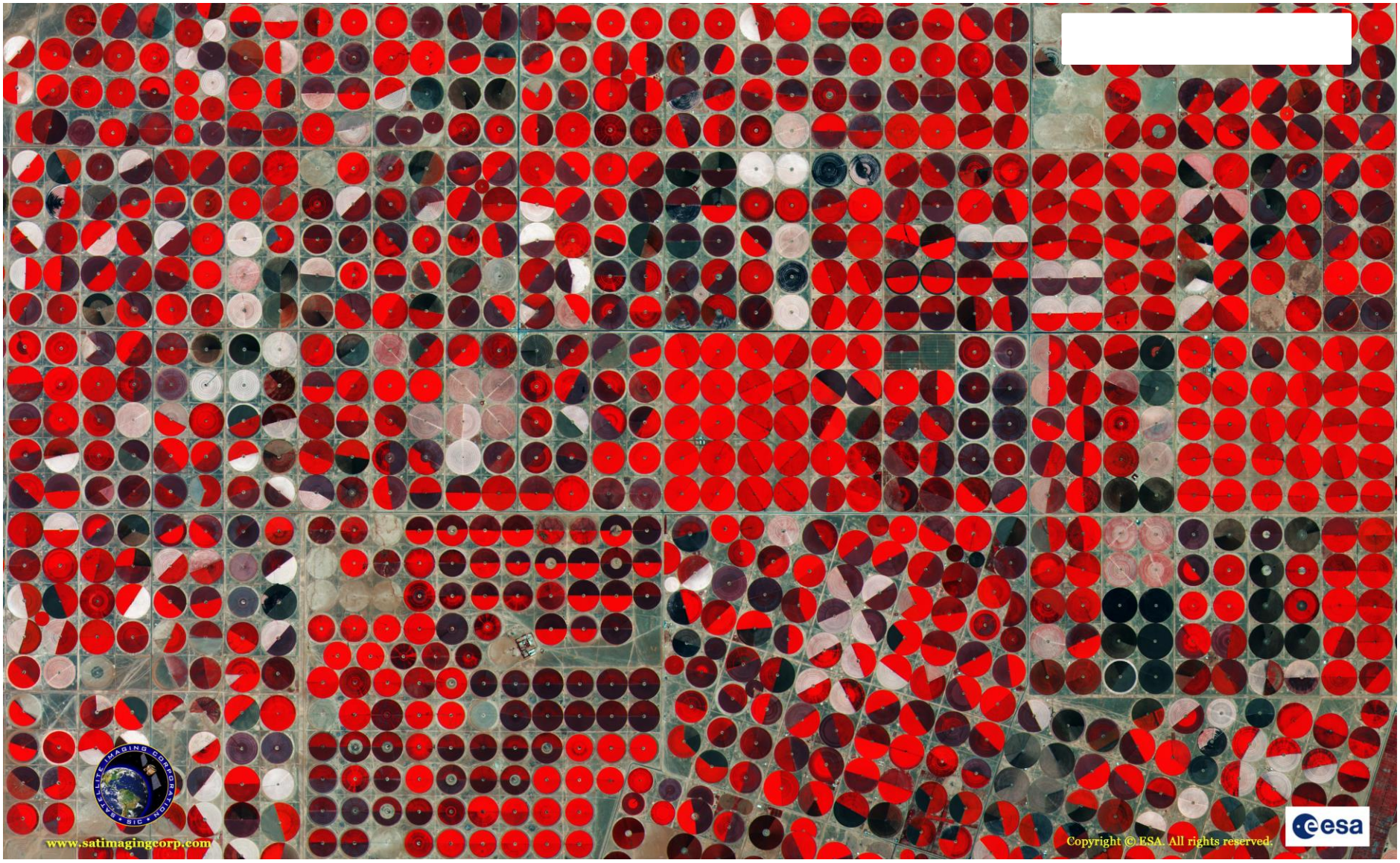
March 2013

LANDSAT 9

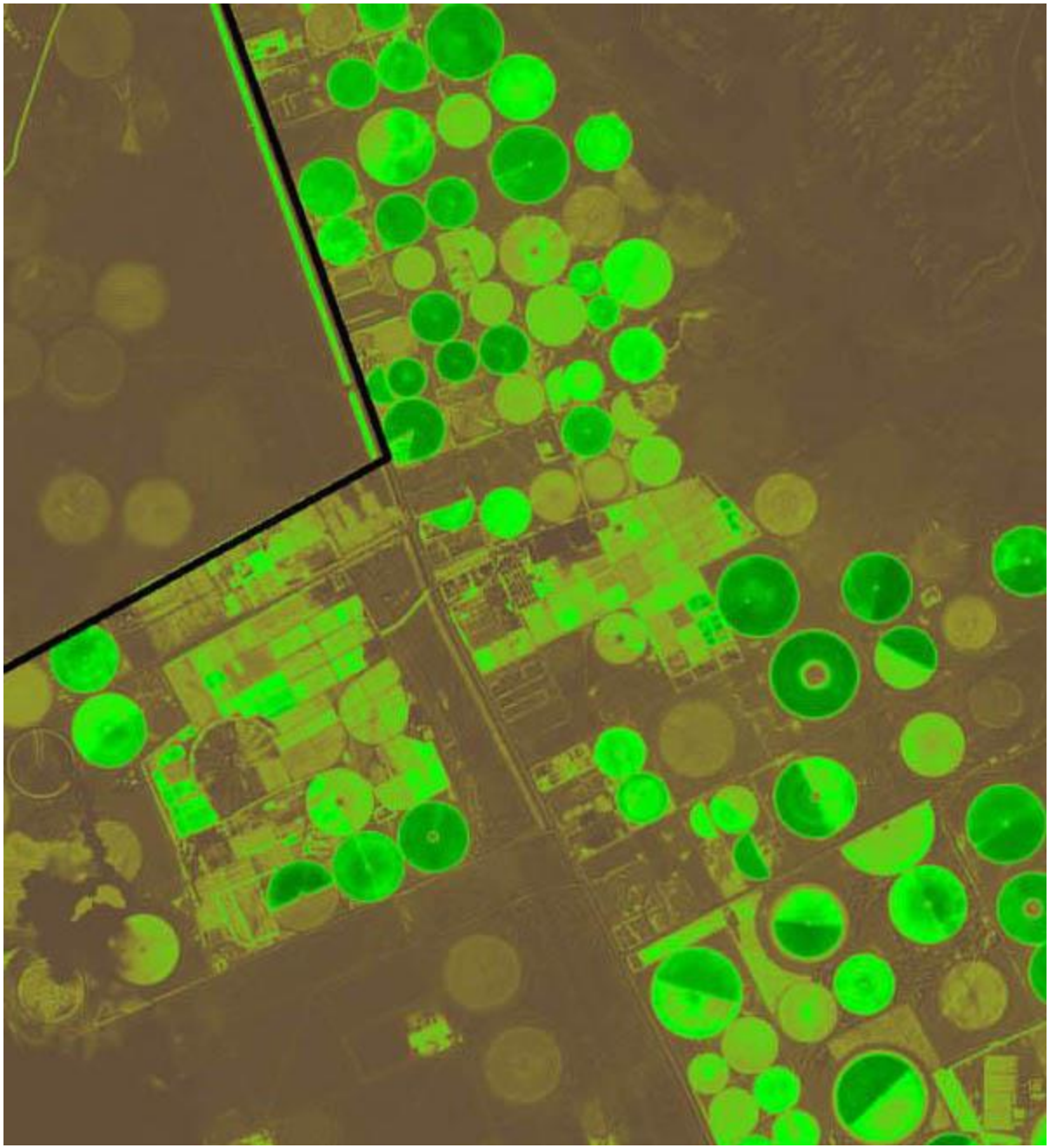
(Landsat Data Continuity Mission (LDCM))

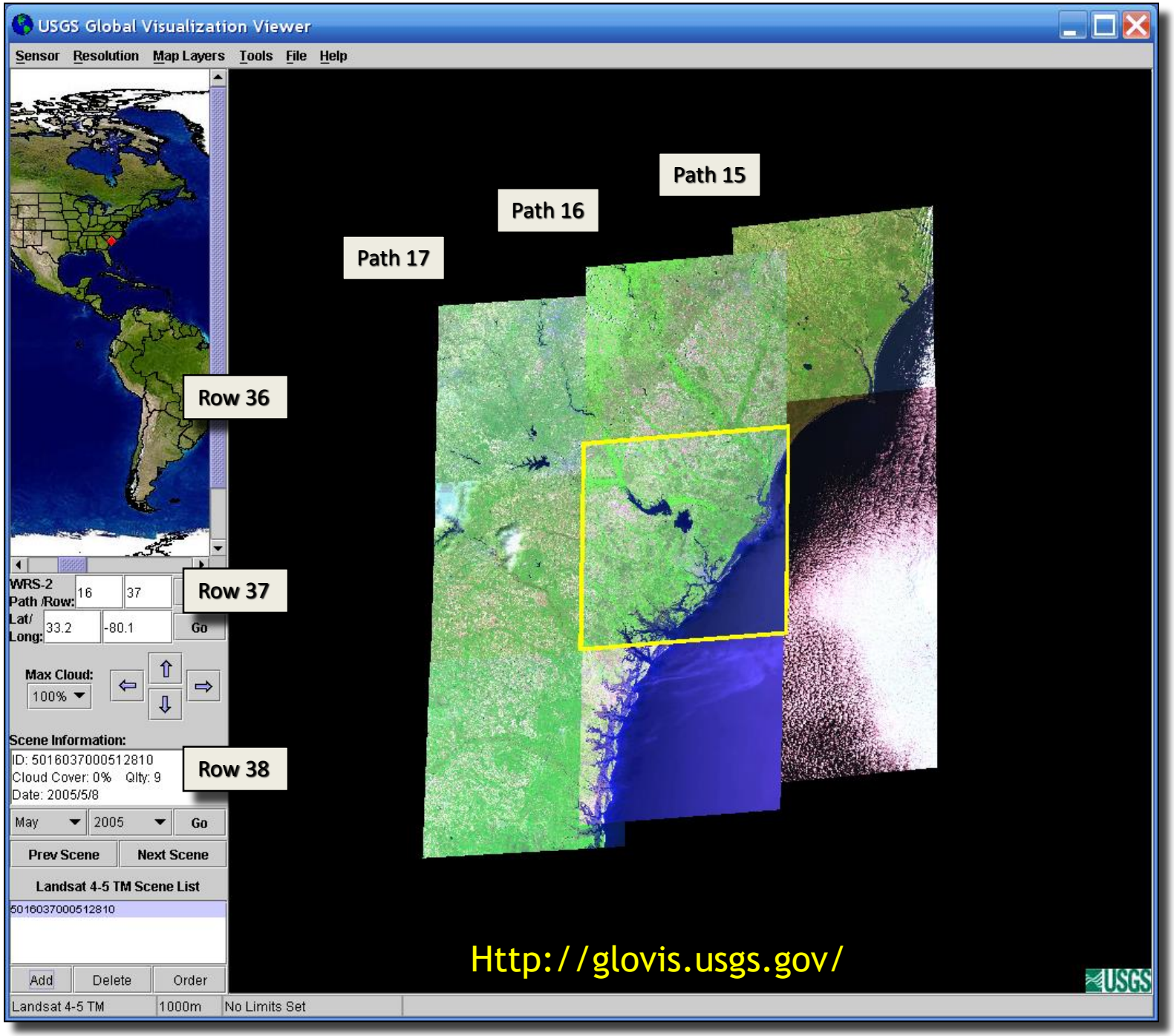
- Launched on September 27, 2021 (to replace Landsat7)
- Operational Land Imager (OLI-2) and the Thermal Infra Red Sensor (TIRS-2)
- Resolution
  - Panchromatic - 15 M
  - Multispectral - 30 M
  - Thermal - 100 m
- Equator Crossing- 10:00 AM
- 12-bit quantization of data

Can you guess what the below pattern is?



LANDSAT-8




USGS Global Visualization Viewer
Sensor
Resolution
Map Layers
Tools
File
Help
Path 15
Path 16
Path 17
Row 36
Row 37
Row 38
WRS-2 Path /Row:
16
37
Lat/ Long:
33.2
-80.1
Go
Max Cloud:
100%
Scene Information:
ID: 5016037000512810
Cloud Cover: 0% Qlty: 9
Date: 2005/5/8
May
2005
Go
Prev Scene
Next Scene
Landsat 4-5 TM Scene List
5016037000512810
Add
Delete
Order
Landsat 4-5 TM
1000m
No Limits Set
Http://glovis.usgs.gov/
USGS